

2

PLANE SURFACES AND PRISMS

The behavior of a beam of light upon reflection or refraction at a plane surface is of basic importance in geometrical optics. Its study will reveal several of the features that will later have to be considered in the more difficult case of a curved surface. Plane surfaces often occur in nature, e.g., as the cleavage surfaces of crystals or as the surfaces of liquids. Artificial plane surfaces are used in optical instruments to bring about deviations or lateral displacements of rays as well as to break light into its colors. The most important devices of this type are prisms, but before taking up this case of two surfaces inclined to each other, we must examine rather thoroughly what happens at a single plane surface.

2.1 PARALLEL BEAM

The parallel beam remains parallel after reflection or refraction at a plane surface, as shown in Fig. 2A(a). Refraction causes a change in width of the beam which is easily seen to be in the ratio $(\cos \phi')/(\cos \phi)$, whereas the reflected beam remains of the same width. There is also chromatic dispersion of the refracted beam but not of the reflected one.

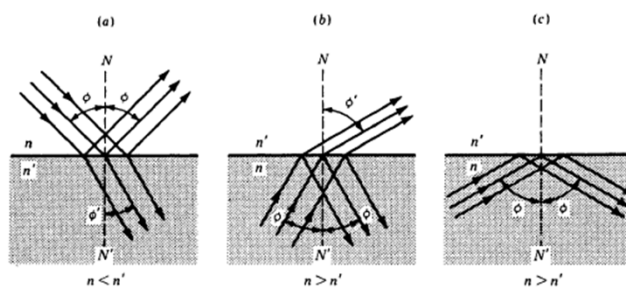


FIGURE 2A

Reflection and refraction of a parallel beam: (a) external reflection; (b) internal reflection at an angle smaller than the critical angle; (c) total reflection at or greater than the critical angle.

Reflection at a surface where n increases, as in Fig. 2A(a), is called *external reflection*. It is also frequently termed *rare-to-dense* reflection because the relative magnitudes of n correspond roughly (though not exactly) to those of the actual densities of materials. In Fig. 2A(b) is shown a case of *internal reflection* or *dense-to-rare* reflection. In this particular case the refracted beam is narrow because ϕ' is close to 90° .

2.2 THE CRITICAL ANGLE AND TOTAL REFLECTION

The critical angle for the boundary separating two optical media is defined as the smallest angle of incidence, in the medium of greater index, for which light is totally reflected.

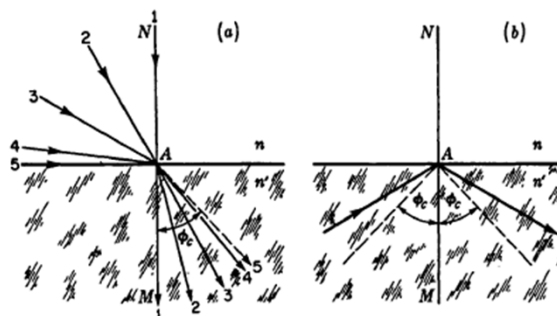


FIGURE 2B

Refraction and total reflection: (a) the critical angle is the limiting angle of refraction; (b) total reflection beyond the critical angle.

Total reflection is really total in the sense that no energy is lost upon reflection.

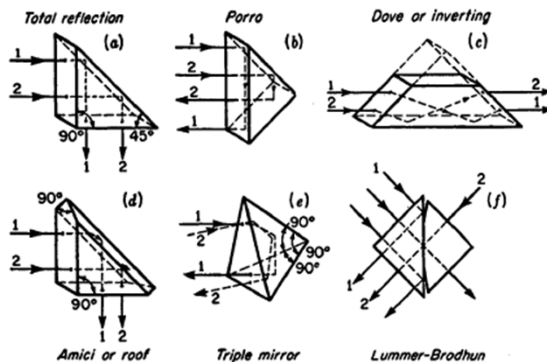


FIGURE 2C

Reflecting prisms utilizing the principle of total reflection.

Since, in the examples shown, the angles of incidence can be as small as 45° , it is essential that this exceed the critical angle in order that the reflection be total. Supposing the second medium to be air ($n' = 1$), this requirement sets a lower limit on the value of the index n of the prism. By Eq. (2a) we must have

$$\frac{n'}{n} = \frac{1}{n} \geq \sin 45^\circ$$

so that $n \geq \sqrt{2} = 1.414$. This condition always holds for glass and is even fulfilled for optical materials having low refractive indices such as Lucite ($n = 1.49$) and fused quartz ($n = 1.46$).

2.3 PLANE-PARALLEL PLATE

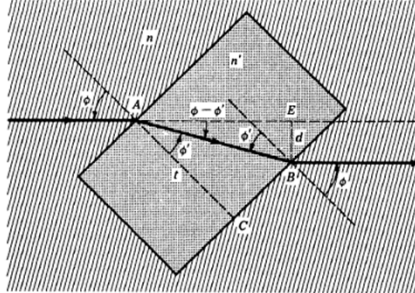


FIGURE 2E
Refraction by a plane-parallel plate.

we may apply the law of refraction and some simple trigonometry to find the displacement d . Starting with the right triangle ABE , we can write

$$d = l \sin (\phi - \phi') \quad (2b)$$

$$d = l(\sin \phi \cos \phi' - \sin \phi' \cos \phi) \quad (2c)$$

From the right triangle ABC we can write

$$l = \frac{t}{\cos \phi'}$$

which, substituted in Eq. (2c), gives

$$d = t \left(\frac{\sin \phi \cos \phi'}{\cos \phi'} - \frac{\sin \phi' \cos \phi}{\cos \phi'} \right) \quad (2d)$$

$$\sin \phi' = \frac{n}{n'} \sin \phi$$

$$d = t \left(\sin \phi - \frac{\cos \phi}{\cos \phi'} \frac{n}{n'} \sin \phi \right)$$

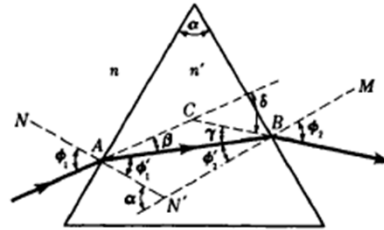
$$d = t \sin \phi \left(1 - \frac{n \cos \phi}{n' \cos \phi'} \right) \quad (2e)$$

2.4 REFRACTION BY A PRISM

In a prism the two surfaces are inclined at some angle α so that the deviation produced by the first surface is not annulled by the second but is further increased. The chromatic dispersion (Sec. 1.10) is also increased, and this is usually the main function of a prism. First let us consider, however, the geometrical optics of the prism for light of a single color, i.e., for *monochromatic light* such as is obtained from a sodium arc.

FIGURE 2F

The geometry associated with refraction by a prism.



$$\frac{\sin \phi_1}{\sin \phi'_1} = \frac{n'}{n} = \frac{\sin \phi_2}{\sin \phi'_2} \quad (2f)$$

The angle of deviation produced by the first surface is $\beta = \phi_1 - \phi'_1$,

by the second surface is $\gamma = \phi_2 - \phi'_2$.

The total angle of deviation δ

$$\delta = \beta + \gamma \quad (2g)$$

Since NN' and MN' are perpendicular to the two prism faces, α is also the angle at N' . From triangle ABN' and the exterior angle α , we obtain

$$\alpha = \phi'_1 + \phi'_2 \quad (2h)$$

Combining the above equations, we obtain

$$\begin{aligned} \delta = \beta + \gamma &= \phi_1 - \phi'_1 + \phi_2 - \phi'_2 = \phi_1 + \phi_2 - (\phi'_1 + \phi'_2) \\ &= \phi_1 + \phi_2 - \alpha \end{aligned} \quad (2i)$$

2.5 MINIMUM DEVIATION

the angle of deviation δ will be observed to decrease, reach a minimum, and then increase again, as shown in Fig. 2G.

The smallest deviation angle, called the *angle of minimum deviation* δ_m ,

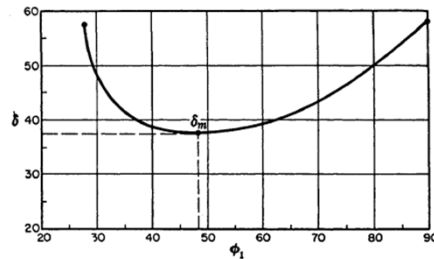


FIGURE 2G
A graph of the deviation produced by a 60° glass prism of index $n' = 1.50$.
At minimum deviation $\delta_m = 37.2^\circ$, $\phi_1 = 48.6^\circ$, and $\phi'_1 = 30.0^\circ$.

(see Fig. 2H). In this special case

$$\phi_1 = \phi_2 \quad \phi'_1 = \phi'_2 \quad \beta = \gamma \quad (2j)$$

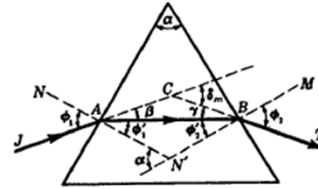


FIGURE 2H
The geometry of a light ray traversing a prism at minimum deviation.

In the triangle ABC in Fig. 2H the exterior angle δ_m equals the sum of the opposite interior angles $\beta + \gamma$. Similarly, for the triangle ABN' , the exterior angle α equals the sum $\phi'_1 + \phi'_2$. Consequently

$$\alpha = 2\phi'_1 \quad \delta_m = 2\beta \quad \phi_1 = \phi'_1 + \beta$$

Solving these three equations for ϕ'_1 and ϕ_1 gives

$$\phi'_1 = \frac{1}{2}\alpha \quad \phi_1 = \frac{1}{2}(\alpha + \delta_m)$$

Since by Snell's law $n'/n = (\sin \phi_1)/(\sin \phi'_1)$,

$$\frac{n'}{n} = \frac{\sin \frac{1}{2}(\alpha + \delta_m)}{\sin \frac{1}{2}\alpha} \quad (2k)$$

The most accurate measurements of refractive index are made by placing the sample in the form of a prism on the table of a spectrometer and measuring the angles δ_m and α , the former for each color desired. When prisms are used in spectroscopes and spectrographs, they are always set as nearly as possible at minimum deviation because otherwise any slight divergence or convergence of the incident light would cause astigmatism in the image.

2.6 THIN PRISMS

$$\frac{n'}{n} = \frac{\sin \frac{1}{2}(\delta_m + \alpha)}{\sin \frac{1}{2}\alpha} = \frac{\delta_m + \alpha}{\alpha}$$

$$\delta = (n' - 1)\alpha \quad (21)$$

Thin prism in air

It is customary to measure the *power* of a prism by the deflection of the ray in centimeters at a distance of 1 m, in which case the unit of power is called the *prism diopter* (D). A prism having a power of 1 prism diopter therefore displaces the ray on a screen 1 m away by 1 cm. In Fig. 21(a) the deflection on the screen is x cm and is numerically equal to the power of the prism. For small values of δ it will be seen that the power in prism diopters is essentially the angle of deviation δ measured in units of 0.01 rad, or 0.573° .

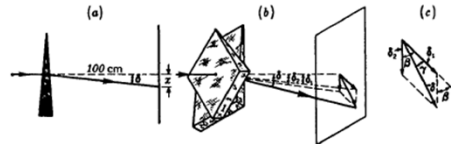


FIGURE 21
Thin prisms: (a) the displacement x in centimeters at a distance of 1 m gives the power of the prism in diopters; (b) Risley prism of variable power; (c) vector addition of prism deviations.

2.7 COMBINATIONS OF THIN PRISMS

In measuring binocular accommodation, ophthalmologists make use of a combination of two thin prisms of equal power which can be rotated in opposite directions in their own plane [Fig. 21(b)]. Such a device, known as the *Risley* or *Herschel prism*, is equivalent to a single prism of variable power. When the prisms are parallel, the power is twice that of either one; when they are opposed, the power is zero. To

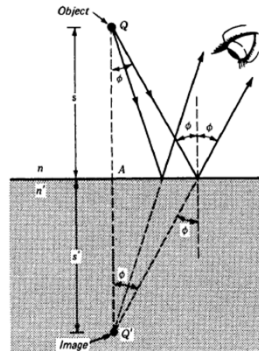
the deviations add vectorially.

the resultant deviation δ will in general be, from the law of cosines,

$$\delta = \sqrt{\delta_1^2 + \delta_2^2 + 2\delta_1\delta_2 \cos \beta} \quad (2m)$$

where β is the angle between the two prisms.

2.10 REFLECTION OF DIVERGENT RAYS



The point Q' is said to be a *virtual image* of Q .

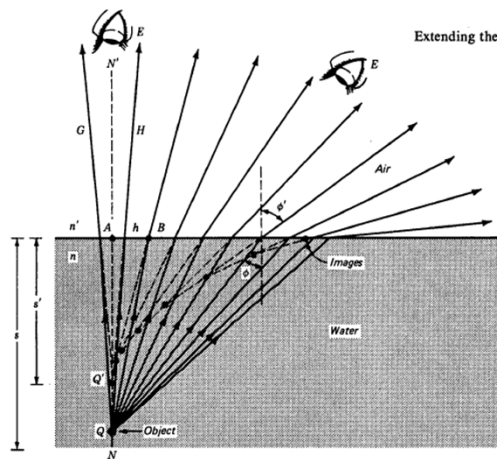
FIGURE 2M
The reflection of divergent rays of light from a plane surface.

When a divergent pencil of light is reflected at a plane surface, it remains divergent. All rays originating from a point Q (Fig. 2M) will after reflection appear to come from another point Q' symmetrically placed behind the mirror. The proof of this proposition follows at once from the application of the law of reflection [Eq. (1j)], according to which all the angles labeled ϕ in the figure must be equal. Under these conditions the distances QA and AQ' along the line QAQ' drawn perpendicular to the surface must be equal; i.e.,

$$s = s'$$

object distance = image distance

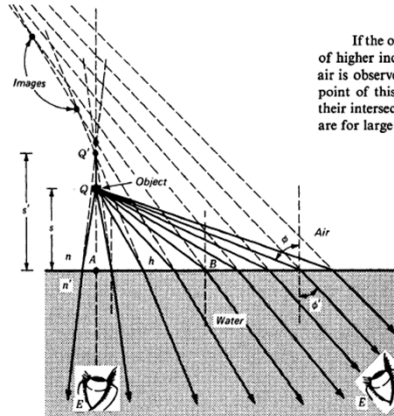
2.11 REFRACTION OF DIVERGENT RAYS



Extending these emergent rays backward, we locate their intersections

These are image points, or *virtual images*.

FIGURE 2N
Image positions of an object under water as seen by an observer above; $n > n'$.



If the object is located in the less dense medium and is observed from the medium of higher index, we obtain an entirely different view (see Fig. 20). An object Q in air is observed by an underwater swimmer or fish. Rays of light diverging from any point of this object are refracted according to Snell's law. Extended backward to their intersections, their virtual images are located. Note how far away these images are for large angles of ϕ and ϕ' .

FIGURE 20
Image positions of an object in air as seen by an observer under water; $n < n'$.

2.12 IMAGES FORMED BY PARAXIAL RAYS

Rays for which angles are small enough to permit setting the cosines equal to unity and the sines and tangents equal to the angles are called *paraxial rays*.

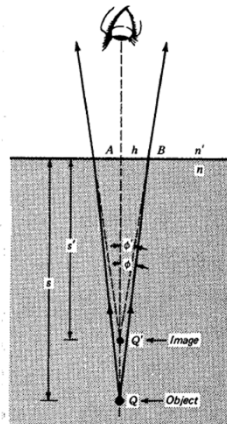


FIGURE 2P
Paraxial rays for an object in water and observed from the air above.

Consider the right triangles QAB and $Q'AB$

$$h = s \tan \phi = s' \tan \phi'$$

$$s' = s \frac{\tan \phi}{\tan \phi'} = s \frac{\sin \phi \cos \phi'}{\cos \phi \sin \phi'}$$

$$s' = s \frac{n' \cos \phi'}{n \cos \phi} \quad (2r) \quad \frac{\sin \phi}{\sin \phi'} = \frac{n'}{n}$$

angles ϕ and ϕ' are very small;

$$s' = s \frac{\phi}{\phi'} \quad \text{or} \quad \frac{s'}{s} = \frac{\phi}{\phi'} \quad (2s)$$

$$\frac{\phi}{\phi'} = \frac{n'}{n} \quad (2t)$$